

Water Jug Problem

Problem Statement

You are given two jugs: one has a capacity of 4 litres and the other has a capacity of 3 litres. Neither jug has any measurement markings. You have an unlimited supply of water.

Your task is to measure exactly **2 litres of water** using only these two jugs.

Allowed Operations

1. Fill either jug completely.
2. Empty either jug completely.
3. Pour water from one jug into the other until the source jug is empty or the destination jug is full.

Question

Find a sequence of operations that leaves exactly 2 litres of water in one of the jugs.

Practice

Try solving the problem yourself first. Record each step as: (4-litre jug, 3-litre jug).